Lightning-Fast
Web Performance

Scott Jehl, WebPageTest Team



Course Agenda

1. Introduction
2. Why Performance Matters
3. **Metrics! How do we define fast?**
4. Identifying Performance Problems
5. Making Things Faster
6. Wrapping Up

Metrics.

How do we define fast?

Fully Loaded

Time to First Byte

Document Complete

First Paint

First Contentful Paint

Page Weight

Speed Index

Time To Interactive



https://developers.google.com/web/updates/2019/02/rendering-on-the-web

User-centric performance metrics

When a user navigates to a web page, they're typically looking for visual feedback to reassure them that everything is going to work as expected.

Is it happening?	Did the navigation start successfully? Has the server responded?
Is it useful?	Has enough content rendered that users can engage with it?
Is it usable?	Can users interact with the page, or is it still busy loading?
Is it delightful?	Are the interactions smooth and natural, free of lag and jank?

To understand when a page delivers this feedback to its users, we've defined several new metrics:

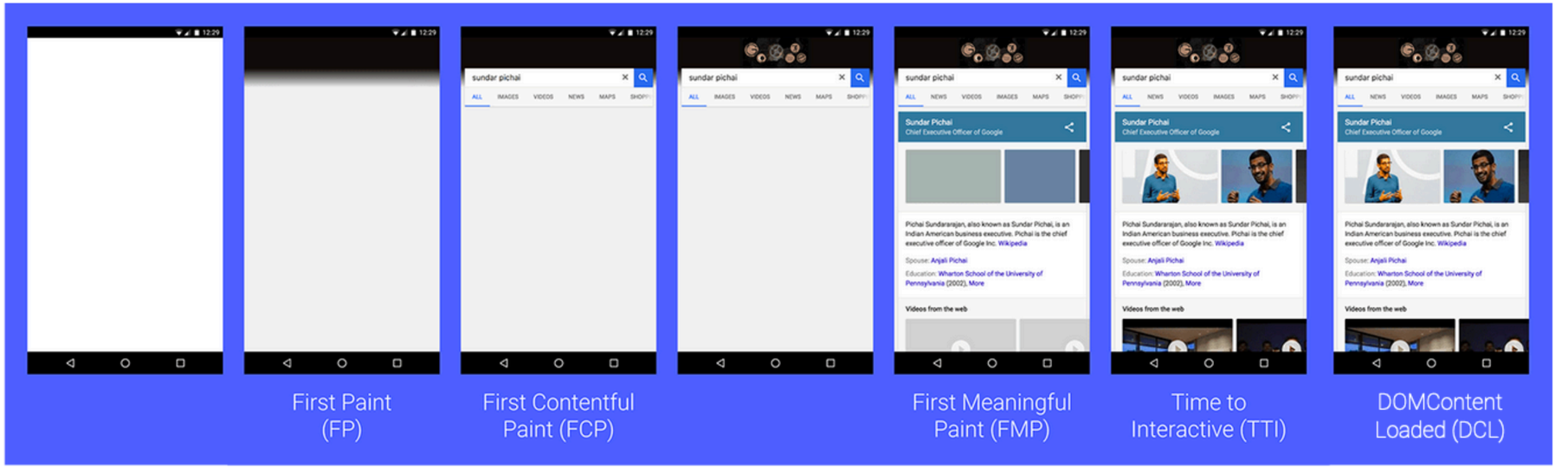


https://developers.google.com/web/fundamentals/performance/user-centric-performance-metrics

"Is it happening?"

"Is it useful?"

"Is it usable?"



Fully (

Visually Complete (VC)

PageSpeed Insight #1

Key UX Factor

Key UX Factor

PageSpeed Insight #2

https://developers.google.com/web/fundamentals/performance/user-centric-performance-metrics

TTFB

Time to First Byte

the time between clicking a link and the first bit of content coming in.

What contributes to TTFB?

- Initial request to a website
- DNS resolution to a to server's IP address
- Establish connection with the server
- Secure connection with the server
- Potential redirects
- Response

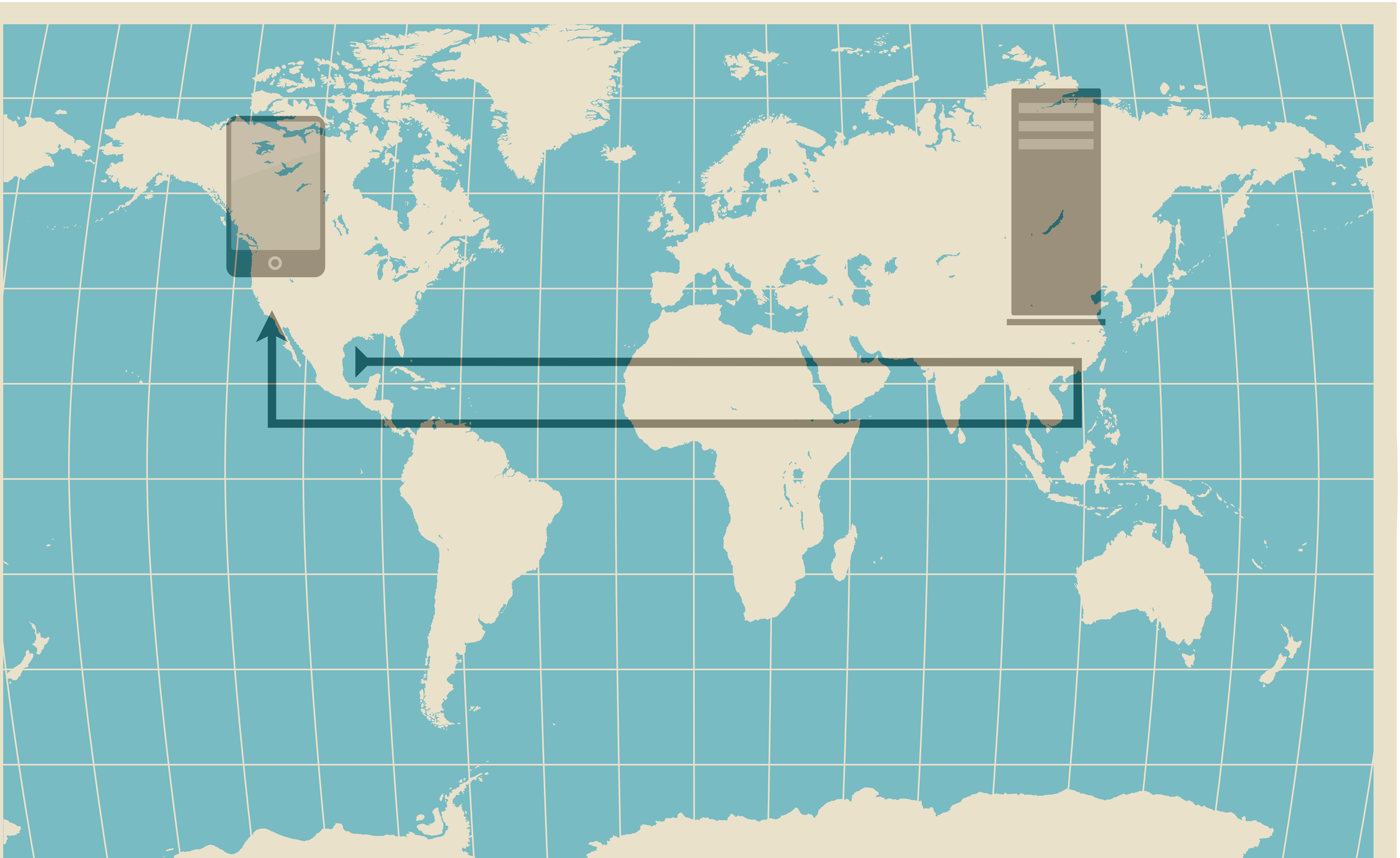
“

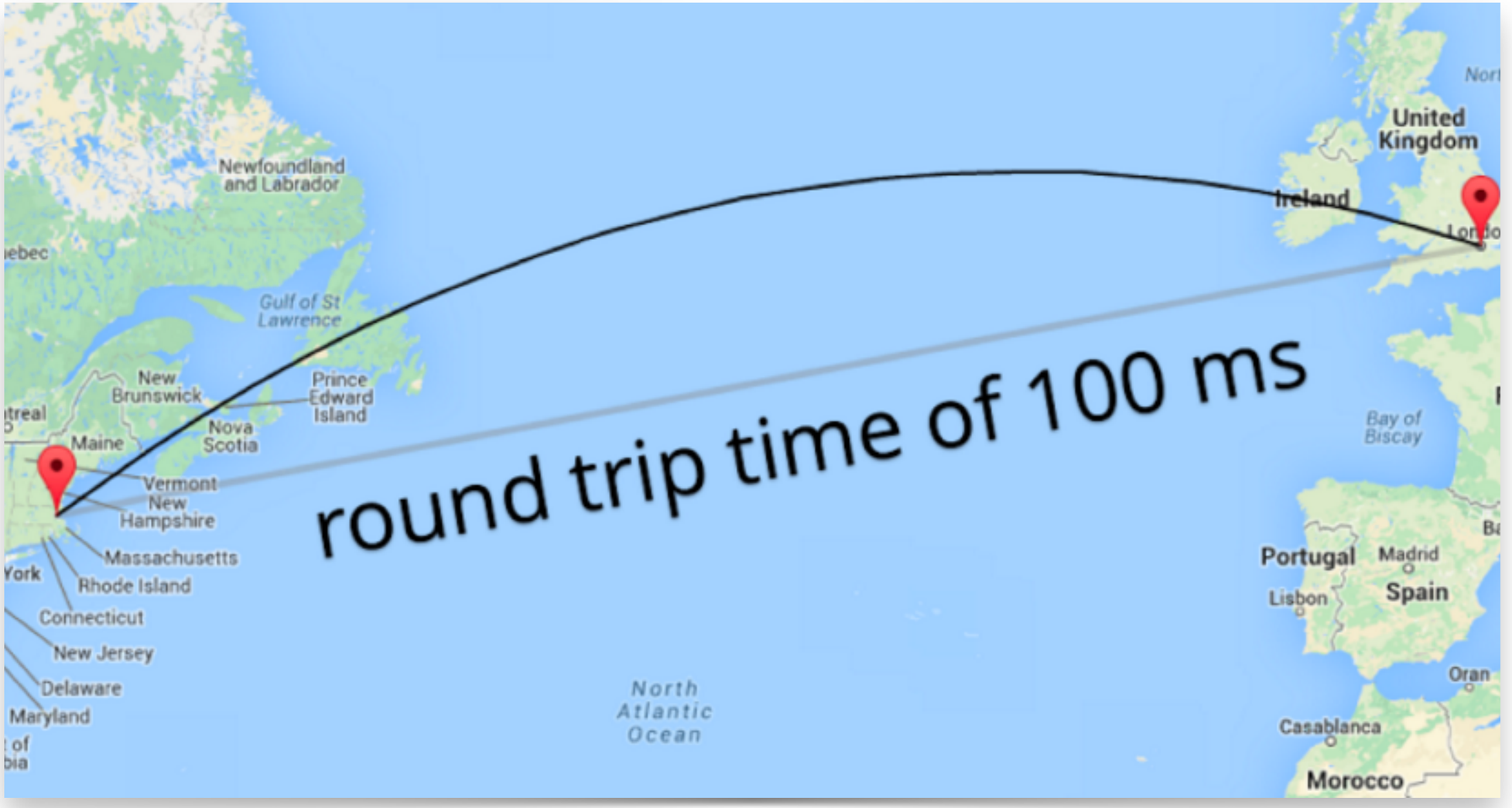
when it comes to your web browsing experience, it turns out that **latency, not bandwidth,** is likely the constraining factor today.

Ilya Grigorik



https://www.igvita.com/2012/07/19/latency-the-new-web-performance-bottleneck/

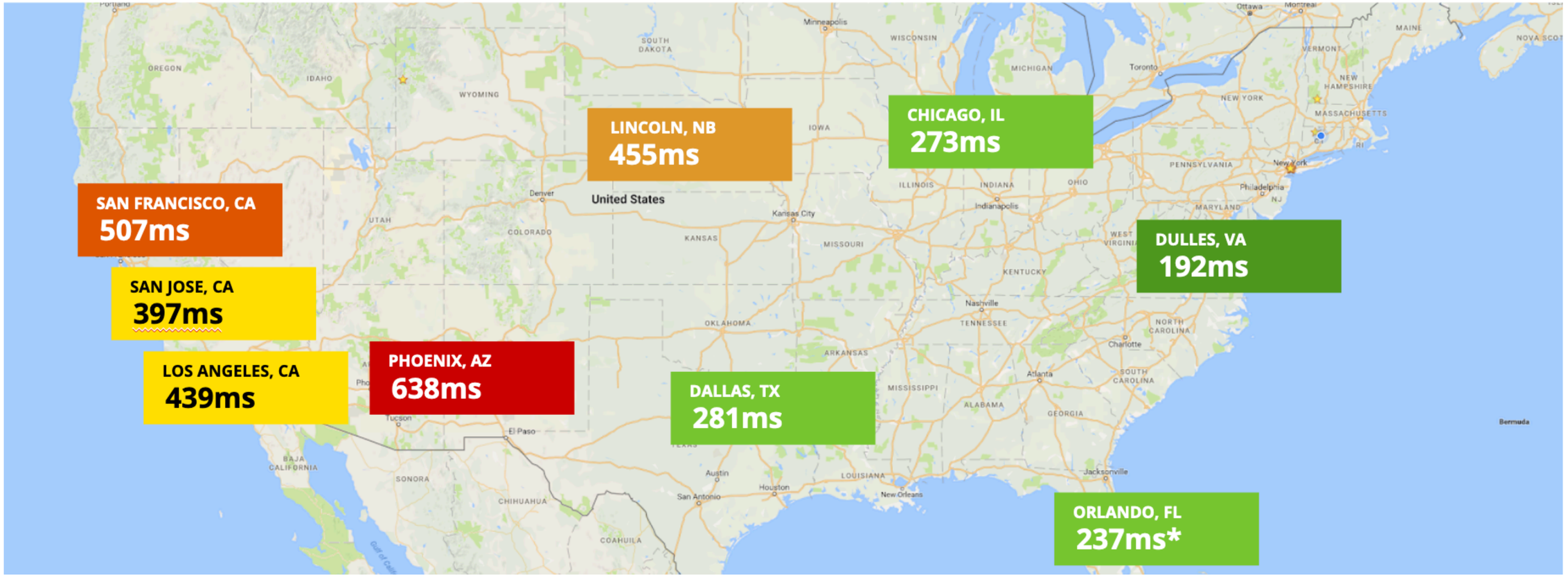




https://ma.ttias.be/googles-quic-protocol-moving-web-tcp-udp/

Baseline HTML: Cable TTFB Timing

Problem: Server round trip time varies by geography



```
Scott:fgwebsite scottjehl$ ping filamentgroup.com
PING filamentgroup.com (45.55.52.167): 56 data bytes
64 bytes from 45.55.52.167: icmp_seq=0 ttl=54 time=114.374 ms
64 bytes from 45.55.52.167: icmp_seq=1 ttl=54 time=85.538 ms
64 bytes from 45.55.52.167: icmp_seq=2 ttl=54 time=94.589 ms
64 bytes from 45.55.52.167: icmp_seq=3 ttl=54 time=85.624 ms
64 bytes from 45.55.52.167: icmp_seq=4 ttl=54 time=84.436 ms
64 bytes from 45.55.52.167: icmp_seq=5 ttl=54 time=84.156 ms
64 bytes from 45.55.52.167: icmp_seq=6 ttl=54 time=84.339 ms
64 bytes from 45.55.52.167: icmp_seq=7 ttl=54 time=84.541 ms
64 bytes from 45.55.52.167: icmp_seq=8 ttl=54 time=84.145 ms
64 bytes from 45.55.52.167: icmp_seq=9 ttl=54 time=85.546 ms
^C
--- filamentgroup.com ping statistics ---
10 packets transmitted, 10 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 84.145/88.729/114.374/9.051 ms
Scott:fgwebsite scottjehl$
```

Server time impacts on TTFB

- Generally, static pages need less server time
- Dynamic pages tend to use more server time

Every 3rd party request:

New connection process

FP

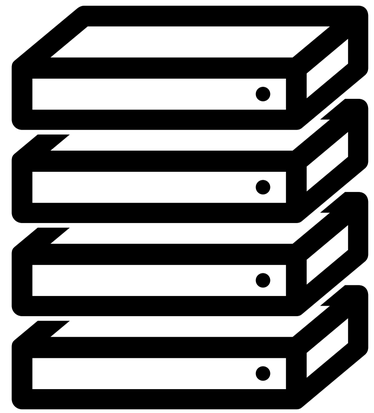
First Paint

The first time pixels start to become visible to the user.



Example.com please!





Sure! Here's the HTML first.

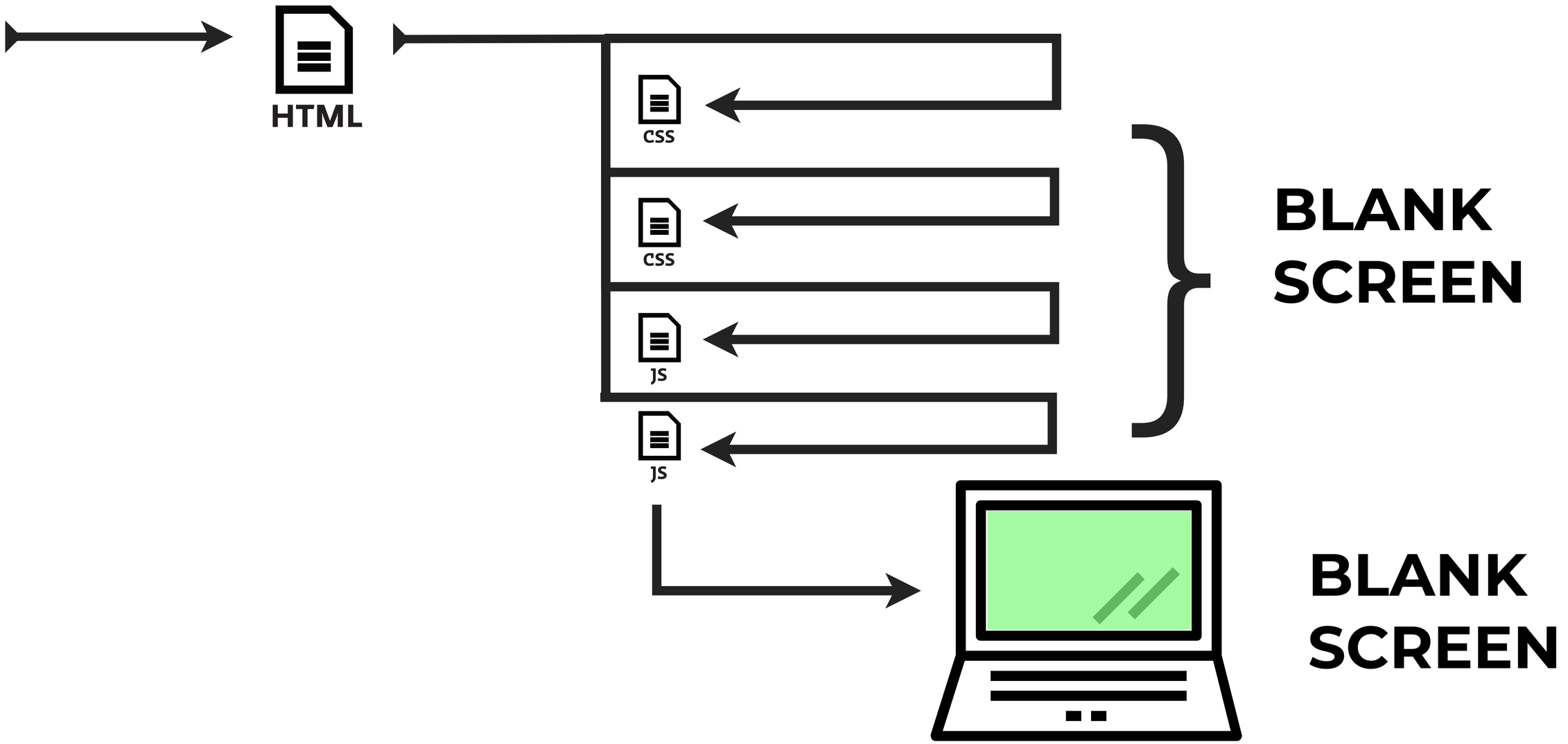
The render blockers

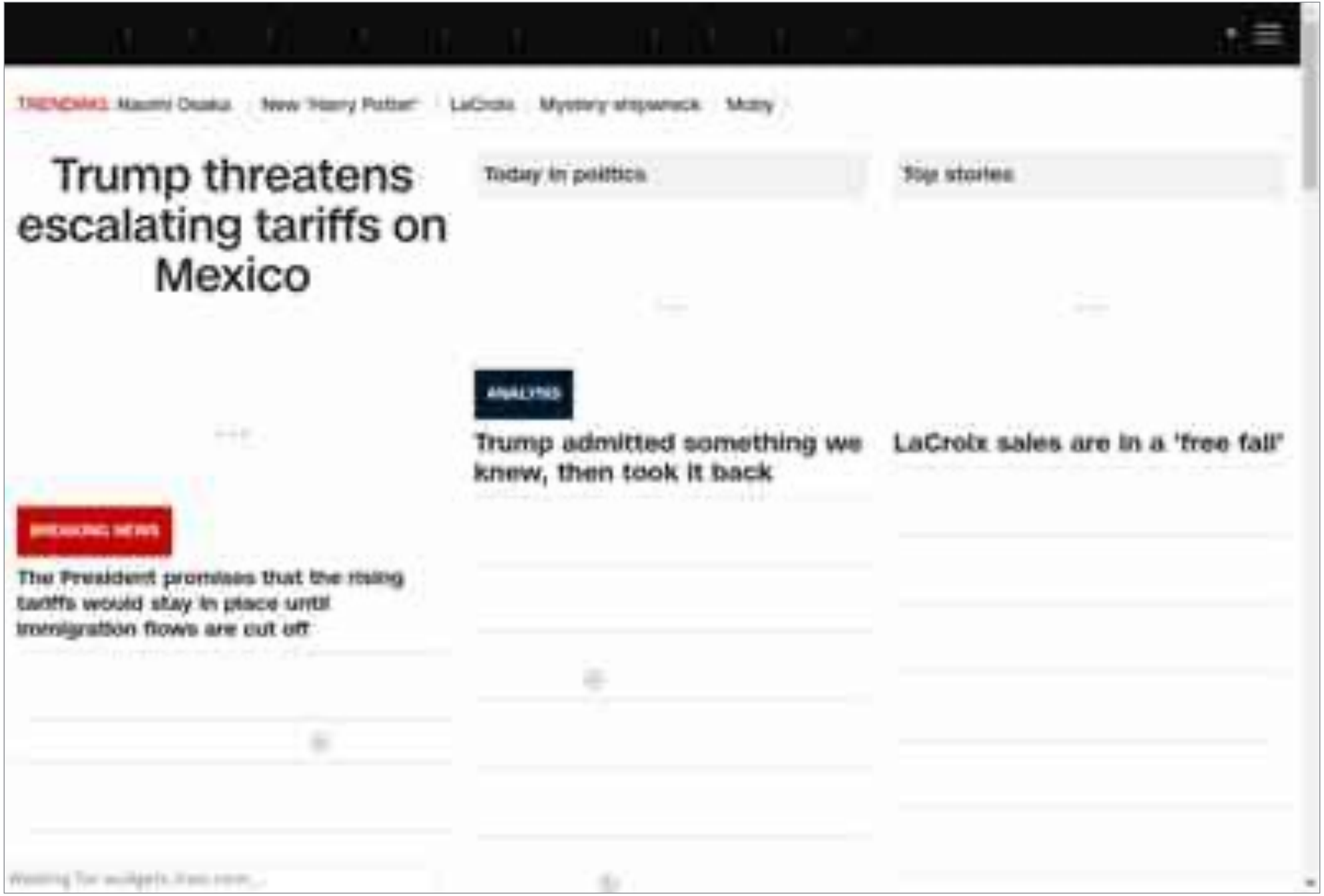
```
<head>
    <link rel="stylesheet" href="site.css">

    <script src="site.js"></script>



</head>
```


HTML
CSS
CSS
JS
JS
BLANK
SCREEN
BLANK
SCREEN


Trump threatens escalating tariffs on Mexico
Today in politics
Top stories
ANALYSIS
Trump admitted something we knew, then took it back
LaCroix sales are in a 'free fall'
BREAKING NEWS
The President promises that the rising tariffs would stay in place until immigration flows are cut off

FMP

First Meaningful Paint

The time when requested content (article body, etc) starts to render meaningfully.

LCP

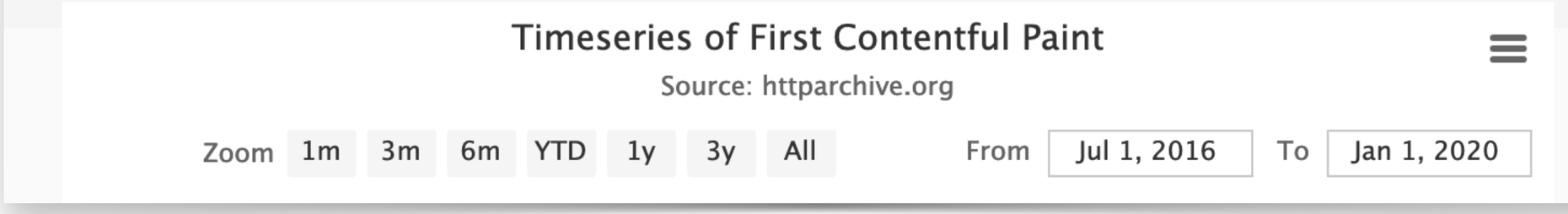
Largest Contentful Paint

The render time of the largest content element visible in the viewport.

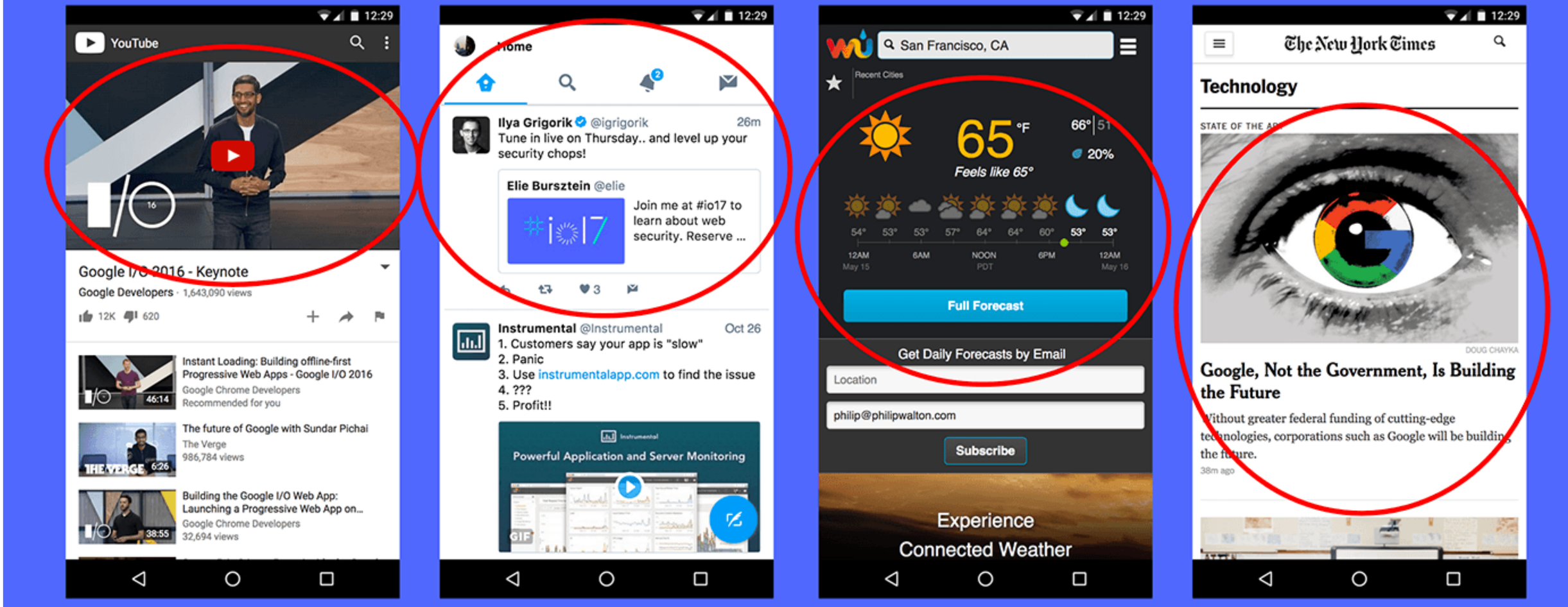
First Contentful Paint

The number of seconds from the time the navigation started until the page's primary content appears on the screen.

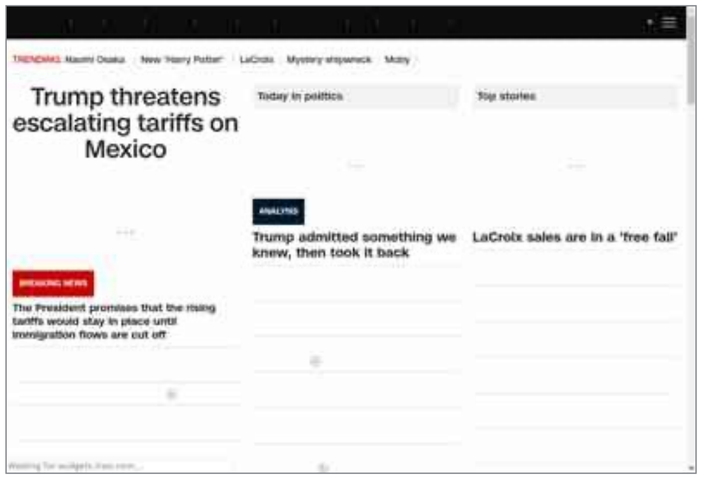
MEDIAN DESKTOP	MEDIAN MOBILE
2.2 seconds	6.4 seconds
▼21.4%	▲33.3%

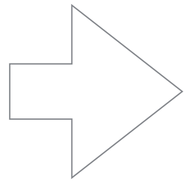


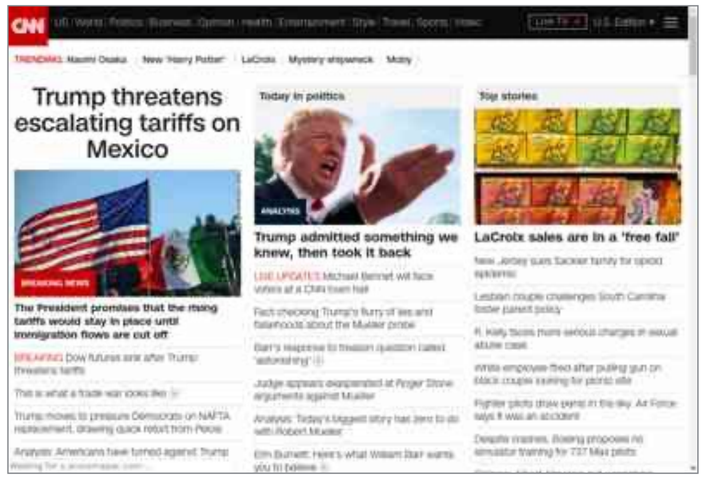
https://httparchive.org/reports/loading-speed#fcp



https://developers.google.com/web/fundamentals/performance/user-centric-performance-metrics







Early Paints

Meaningful Paint

Key Probs: fetching fonts and images sooner, and pre-rendered content

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International Space Station

Nasa to allow commercial access to the International Space Station

Move will allow private astronauts to spend up to 30 days in low-Earth orbit and businesses to shoot film and adverts in space

Jessica Glenza *in New York*

@JessicaGlenza

Fri 7 Jun 2019 12.53 EDT

16



▲ Astronaut Christina Koch: 'Transitioning toward this new model of business is an important step to allow Nasa to move full speed ahead in landing the first woman and the next man on the moon.' Photograph: Handout ./Reuters

Nasa will allow unprecedented commercial access to the International Space Station (ISS) for marketing, business and space tourism, the agency announced on Friday.

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Zuckerberg's Hawaii estate: battle's latest turn 'devastates' local family


Live Djokovic halted by rain after Nadal beats Federer and Konta misses out on final – as it happened


'Straight pride' group removes Brad Pitt as mascot after backlash


Teenagers held over homophobic attack on two women on London bus

Data URIs: often blocky as well.

```
<img src="data:text/
plain; base64,SGVsbG8sIFdvcmxkIQ%3D%3D…">
```


●●○○○ T-Mobile LTE 6:37 AM 96%
Between the Lines | Fast...
m.fastcompany.co...507/between-lines
FAST COMPANY

Your Tuesday Evening Briefing
Here's what you need to know at the end of the day.

Listen to 'The Daily'
The rise of Modi: India's rightward turn.

The Book Review Podcast
The authors Casey Cep and Eliza Griswold discuss their new books.

S&P 500 +0.85% ↑
Dow +0.77% ↑
Nasdaq +1.08% ↑

82°F
85° 69°
Mililani Town, HI

WASHINGTON

Democrats Divided Over Trump's Stonewalling as Calls for Impeachment Inquiry Grow

- Democrats were at odds about how to fight the latest defiance of a House subpoena, this time by the former White House counsel Donald McGahn.
- Some progressive lawmakers are pushing for the first time for an impeachment inquiry, over leaders' objections.

1h ago 841 comments



Donald F. McGahn II, the former White House counsel, skipped a House Judiciary Committee hearing on President Trump's attempts to obstruct the Russia investigation. Erik S Lesser/EPA, via Shutterstock

Trump Expected to Pick Hard-Liner for Immigration Post

The job's duties and title are still being decided, but Ken Cuccinelli, a former Virginia attorney general, is expected to become a policy coordinator.

8h ago 122 comments

```
<script src="path-to-some-advertisement.js"></script>
```



Scott Jehl @scottjehl · 6m

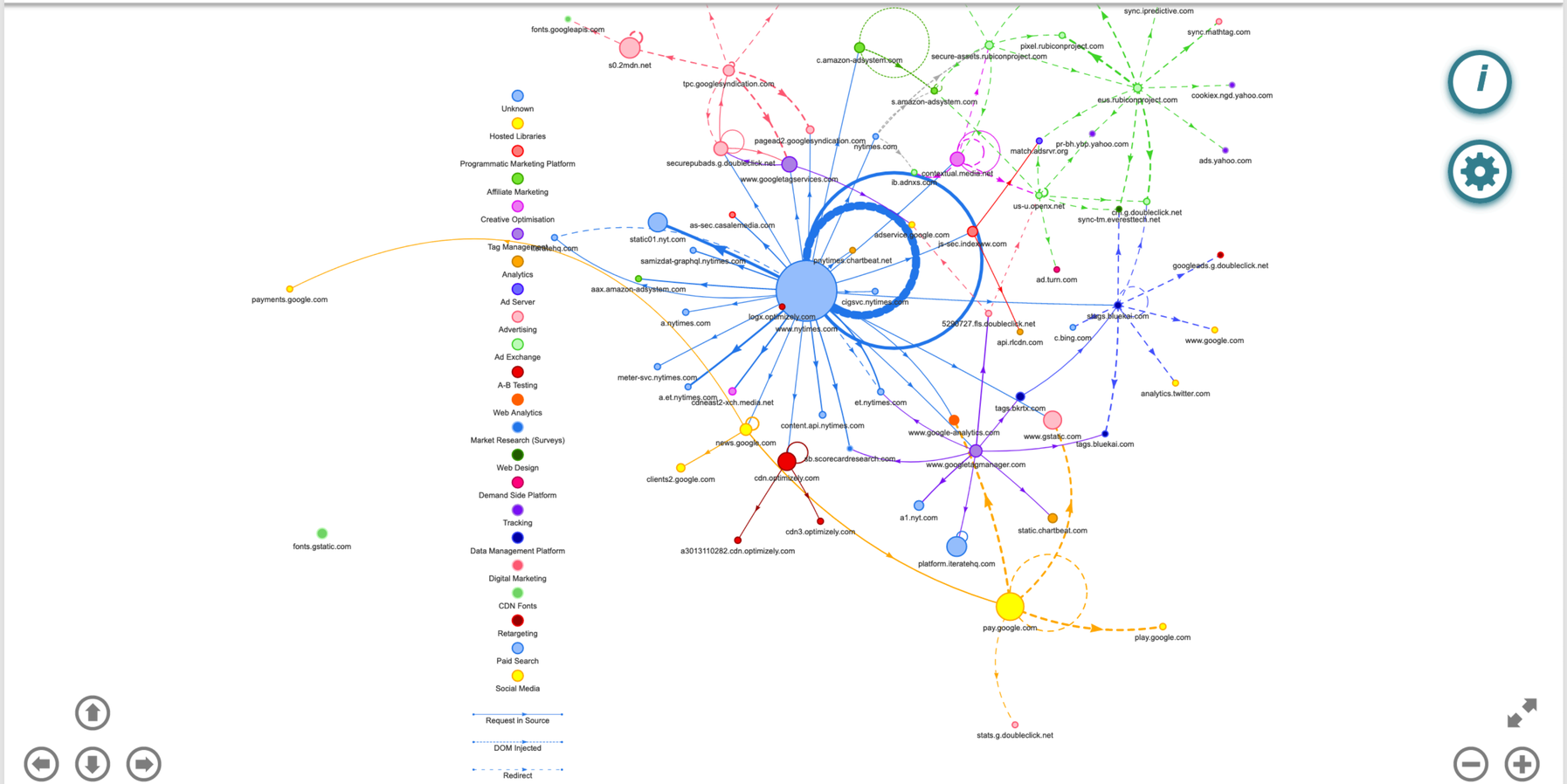
I was curious how long browsers these days will wait for a JavaScript file to load before displaying content that comes after it in the page. So I tested it!

Chrome: 30s
Firefox: 30s
Edge: 20s
IE 11: 7s
Safari: 60s
Android Chrome: 63s
iOS: 75s

Test runs tinyurl.com/swngjl3

https://twitter.com/scottjehl/status/1229275920600649728

Request Map for *http://nytimes.com* on Tuesday the 21st of May at 11pm

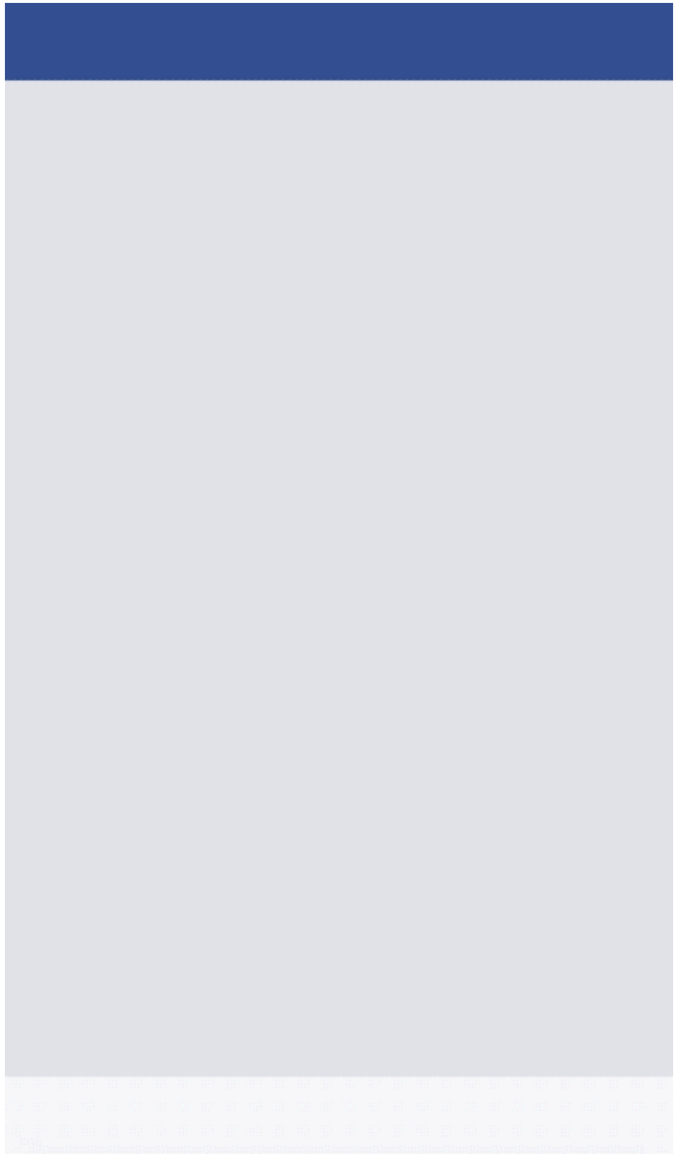


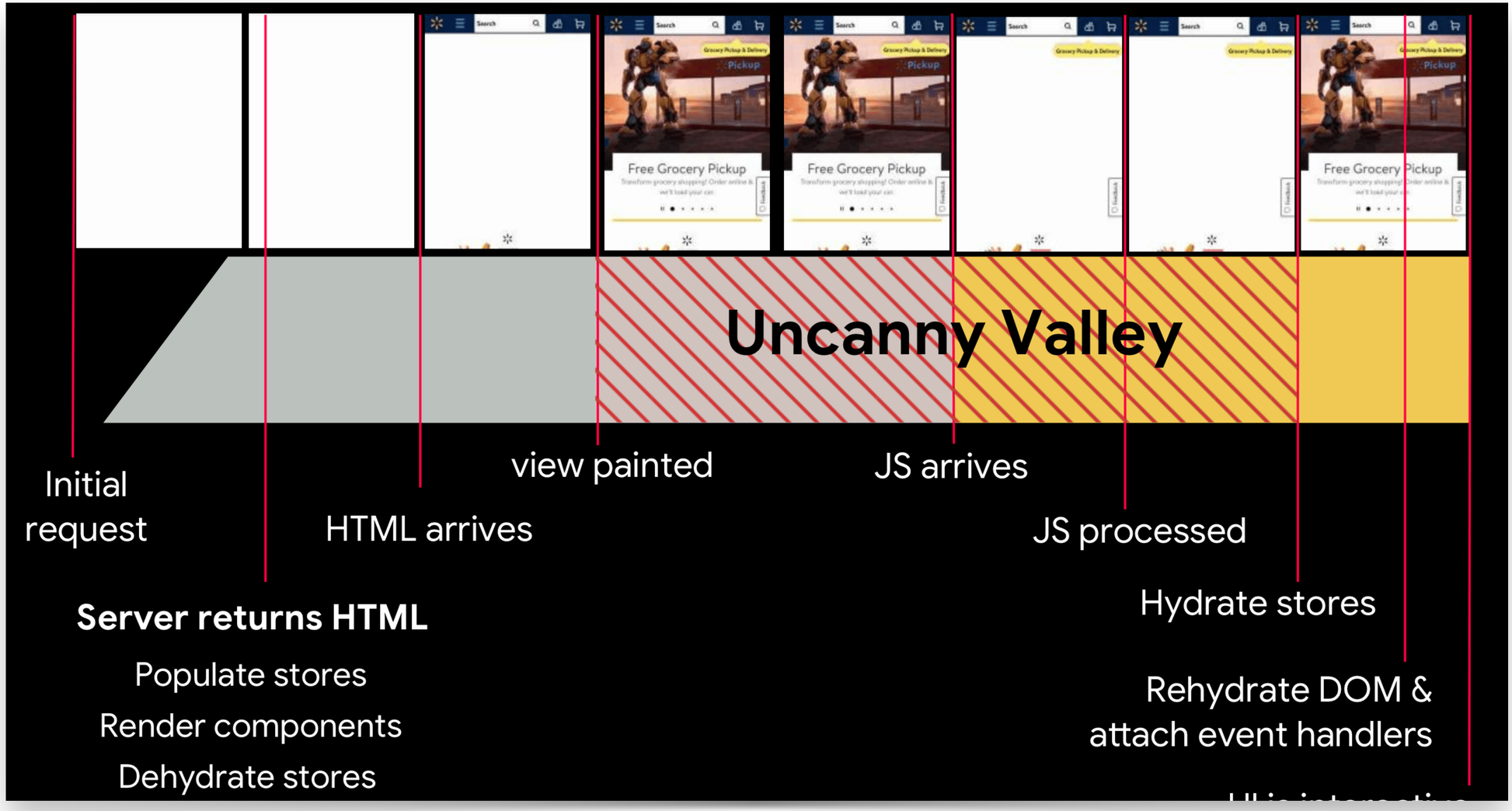
Built by Simon | Download Image | View on webpagetest.org | **View all requests** | Download CSV Data

http://requestmap.webperf.tools/render/190521_J6_a6cad51d836873feba5c4ecc29824feb

Dynamic Scripts can block render

```
document.write('<script src="hello.js"></script>);
```





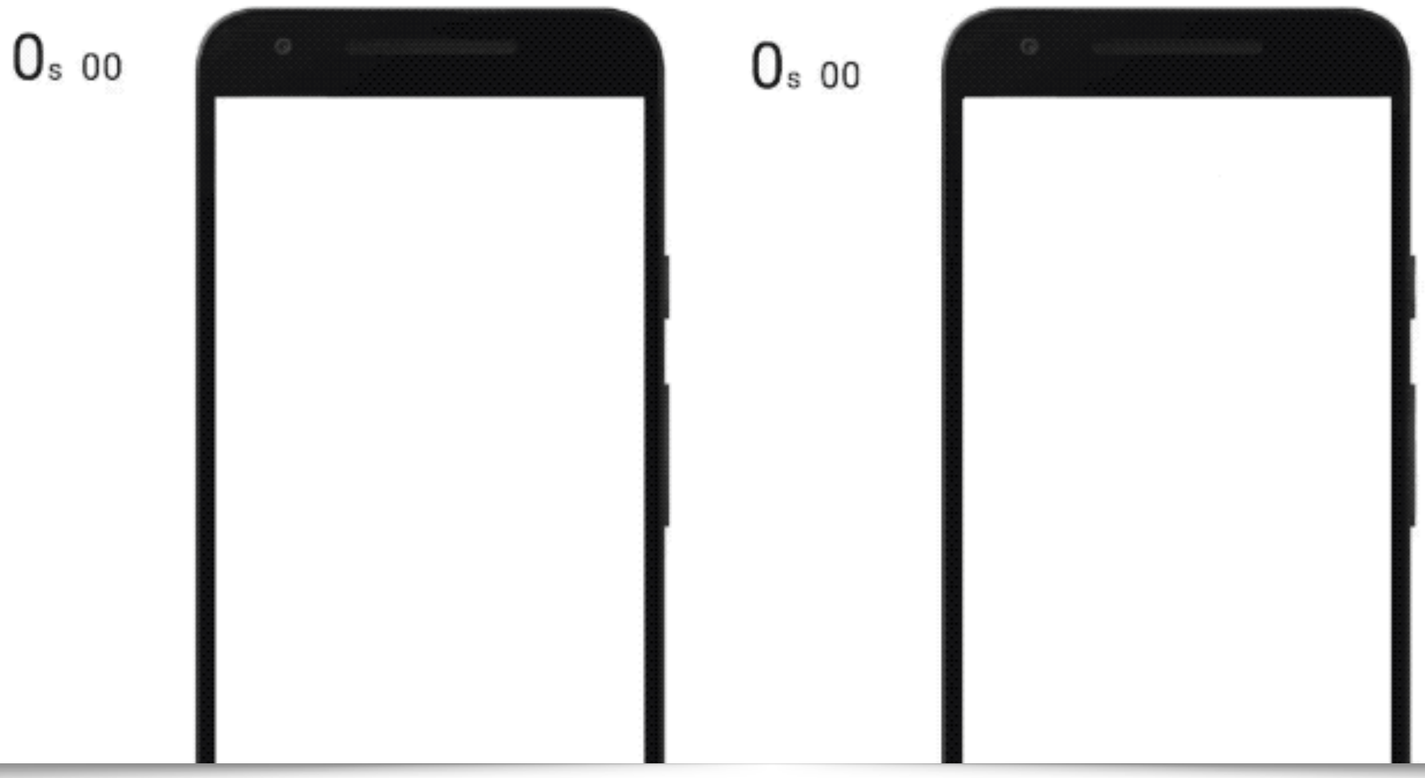
https://addyosmani.com/blog/rehydration/

TTI

Time To Interactive

The time at which a page becomes interactive (events wired up, etc).

TIME TO INTERACTIVE



https://medium.com/@addyosmani/the-cost-of-javascript-in-2018-7d8950fbb5d4

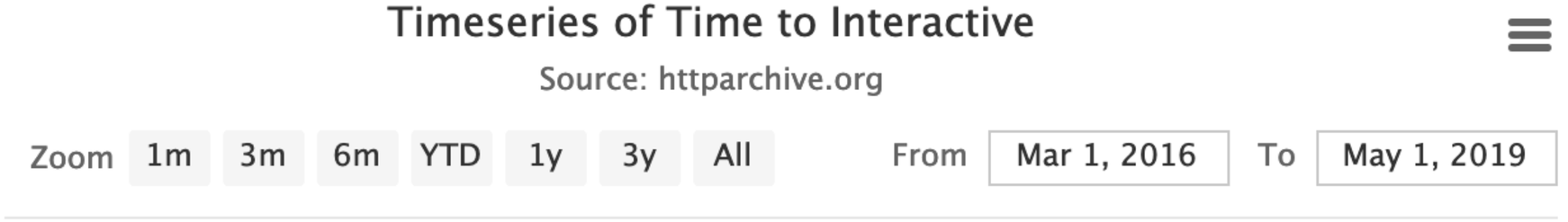
Time to Interactive

The number of seconds from the time the navigation started until the CPU had at least 5 seconds of quiescence. This metric comes from Lighthouse as is only available in mobile tests. Read more.

MEDIAN MOBILE

9.3 seconds

▼23.1%



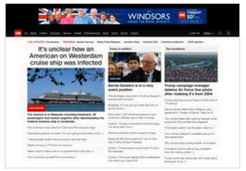
https://httparchive.org/reports/loading-speed#ttci

4

https://www.cnn.com/

0–49 50–89 90–100

Field Data — Over the last 30 days, the field data shows that this page has an **Moderate** speed compared to other pages in the Chrome User Experience Report. We are showing the 75th percentile of FCP and the 95th percentile of FID.



First Contentful Paint (FCP)	2.8 s	First Input Delay (FID)	107 ms
15% 63% 22%		95% 3% 2%	

Show Origin Summary

Lab Data

First Contentful Paint	2.8 s	First Meaningful Paint	3.1 s
Speed Index	6.5 s	First CPU Idle	9.4 s
Time to Interactive	10.7 s	Max Potential First Input Delay	570 ms



https://developers.google.com/speed/pagespeed/insights/?url=cnn.com&tab=desktop



https://www.cnn.com/

0–49 50–89 90–100

Field Data — Over the last 30 days, the field data shows that this page has a **Slow** speed compared to other pages in the Chrome User Experience Report. We are showing the 75th percentile of FCP and the 95th percentile of FID.

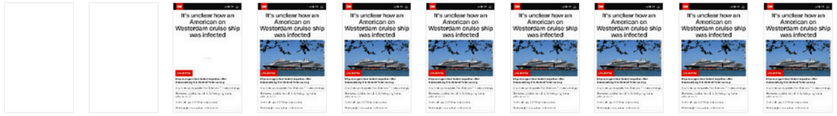
First Contentful Paint (FCP)	4.2 s	First Input Delay (FID)	254 ms
1% 60% 39%		91% 5% 4%	

Show Origin Summary



Lab Data

First Contentful Paint	11.4 s	First Meaningful Paint	12.8 s
Speed Index	25.1 s	First CPU Idle	32.3 s
Time to Interactive	46.3 s	Max Potential First Input Delay	1,500 ms



https://developers.google.com/speed/pagespeed/insights/?url=cnn.com&tab=desktop

“

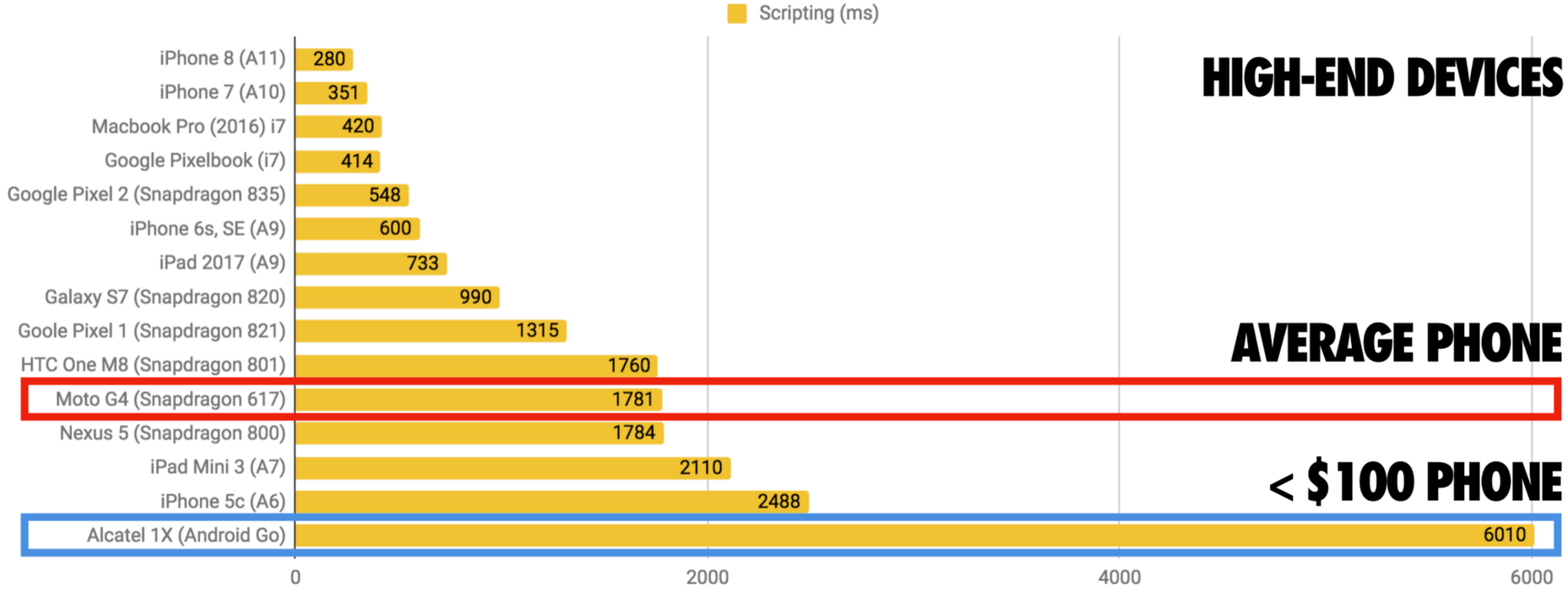
we feel your baseline should be getting interactive in under 5 seconds on a slow 3G connection on a median mobile device.

Addy Osmani



https://medium.com/@addyosmani/the-cost-of-javascript-in-2018-7d8950fbb5d4

2018 JAVASCRIPT PROCESSING TIMES



Tests run during July, 2018 on hardware running the latest versions of Android and iOS available

1MB JS UNCOMPRESSED (200KB min/compressed)

https://medium.com/@addyosmani/the-cost-of-javascript-in-2018-7d8950fbb5d4

“

When creating great user experiences, managing CPU usage is now just as important as fast networks.

Mark Zeman, SpeedCurve



https://speedcurve.com/blog/your-javascript-hurts/

Other

Important Metrics...

First Input Delay... First CPU Idle....
Total blocking time... Cumulative layout shift...

Still important:

Fully Loaded

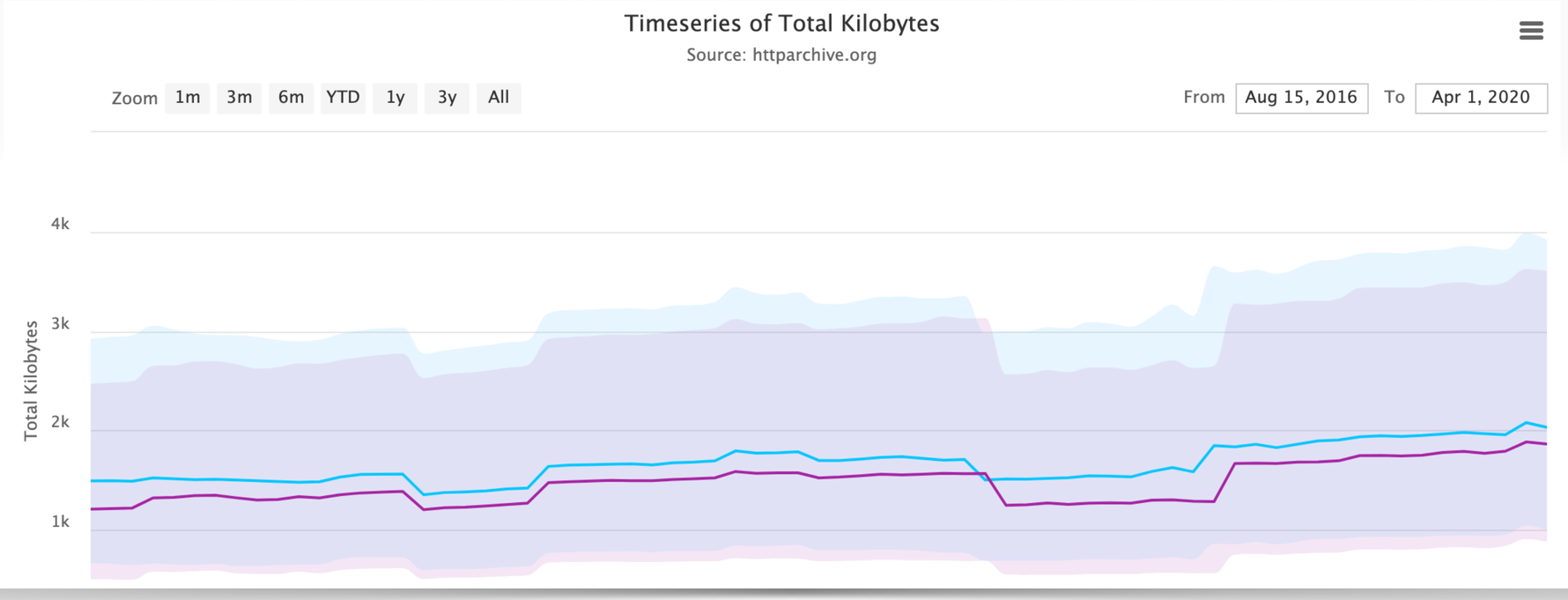
Not a user-perceivable metric (*until they check their data bill!*)

Total Kilobytes

The sum of transfer size kilobytes of all resources requested by the page.

See also: State of the Web

MEDIAN DESKTOP	MEDIAN MOBILE
2031.9 KB	1864.2 KB
▲36.1%	▲54.1%



https://httparchive.org/reports/page-weight

Lightning-Fast
Web Performance

Scott Jehl, WebPageTest Team

